Exercise sheet 2.

Exercise 1

remembering Taylor

$$f(\cancel{x_k+s}) \approx f(\cancel{x_k}) + f'(x_k)s + \frac{1}{2}f''(x_k)s^2$$

$$\frac{d}{ds} \approx 0$$

$$0 = f'(x_k) + f''(x_k)s$$

$$s_k = \frac{-f'(x_k)}{f''(x_k)}$$

$s = x - x_k$

$$x_{k+1} = x_k + s_k$$

multivariable

$$\underbrace{\nabla(x_k+s)}_{0} \approx \nabla f(x_k) + H(x_k).s$$

$x_{k+1} = x_k + s_k$

$$0 \approx \nabla f(x_k) + H(x_k)\, s_k$$

$$s_k = -H(x_k)^{-1}.\nabla f(x_k)$$

$$x_{k+1} = x_k + s_k$$

* Gaus Newton (GN)

$$H = J^T.J$$

* Levenberg - Marguardt (LM)

$$H \approx J^T.J + \mu^2 I$$

$$s_k = -(J^T.J + \mu^2 I)^{-1} J^T F$$

$$(J^T.J + \mu^2 I)s = -J^T F$$

$$\min_{s \in \mathbb{R}^n} \|Js + F\|_2^2 + \mu^2 \|s\|_2^2$$